

CODECHECK certificate 2026-024

10.5281/zenodo.23166593



Item	Value
Title of checked publication	Beyond colour gamuts: Novel metrics for the reproduction of photoreceptor signals
Author(s)	A. C. Hexley  , T. Morimoto  , H. E. Smithson  , M. Spitschan 
Reference	https://doi.org/10.1101/2021.02.27.433203
Codechecker(s)	Simon Christ 
Date of check	29.09.2026
Summary	The code check was straightforward. Only the toolboxes "Statistics and Machine Learning" and "Image Processing" needed to be added in contrast to what was stated in the README. Also the preprint was not up-to-date with the code of the repository but the authors provided the current version upon request.
Repository	https://github.com/takuma929/metrics_photoreceptor_signal_reproduction

Table 1: CODECHECK summary

Output	Comment	Size (b)
codecheck/outputs/fig1a.pdf	Fig 1 - spectral sensitivities 1	39685
codecheck/outputs/fig1b.pdf	Fig 1 - example signals 1	16693
codecheck/outputs/fig1c.pdf	Fig 1 - spectral sensitivities 2	19252
codecheck/outputs/fig1d.pdf	Fig 1 - example signals 2	15798
codecheck/outputs/fig1e_inse rted.pdf	Fig 1 - inset of contrast for R and I	8187
codecheck/outputs/fig1e_inse rted.pdf	Fig 1 - response bar chart 1	8187
codecheck/outputs/fig1e.pdf	Fig 1 - spectral sensitivities 3	8726
codecheck/outputs/fig1g.pdf	Fig 1 - example signals 3	16196
codecheck/outputs/fig1h.pdf	Fig 1 - response bar chart 2	8713
codecheck/outputs/fig2a.pdf	Fig 2 - equal-luminance MacLeod- Boynton locus 1	14703
codecheck/outputs/fig2a_inse rt.pdf	Fig 2 - equal-luminance MacLeod- Boynton locus inset 1	9413
codecheck/outputs/fig2b.pdf	Fig 2 - equal-luminance MacLeod- Boynton locus 2	15351
codecheck/outputs/fig2b_inse rt.pdf	Fig 2 - equal-luminance MacLeod- Boynton locus inset 2	9571
codecheck/outputs/fig2c.pdf	Fig 2 - equal-luminance MacLeod- Boynton locus 3	14189
codecheck/outputs/fig2c_inse rt.pdf	Fig 2 - equal-luminance MacLeod- Boynton locus inset 3	9272
codecheck/outputs/fig3a.pdf	Fig 3a - display primaries	27103
codecheck/outputs/fig3b.pdf	Fig 3b - display primaries	28964
codecheck/outputs/fig3c.pdf	Fig 3c - display primaries	27972
codecheck/outputs/fig3d.pdf	Fig 3d - display primaries	14510
codecheck/outputs/fig3e.pdf	Fig 3e - display primaries	23008
codecheck/outputs/fig3f.pdf	Fig 3f - display primaries	32804
codecheck/outputs/fig3g.pdf	Fig 3g - display primaries	34268
codecheck/outputs/fig3h.pdf	Fig 3h - display primaries	27196
codecheck/outputs/fig4a.png	Fig 4a - chromaticity reproduction diagram	318348
codecheck/outputs/fig4b.pdf	Fig 4b - chromaticity reproduction bar graph	11681
codecheck/outputs/fig4c.pdf	Fig 4c - PSRM (photoreceptor signal reproduction) bar graph	11575
codecheck/outputs/fig5ai.pdf	Fig 5a - PSDM rod and melanopsin distortion 1	138690
codecheck/outputs/fig5aai.pdf	Fig 5a - PSDM rod and melanopsin distortion 2	136340
codecheck/outputs/fig5bi.pdf	Fig 5b - PSDM rod and melanopsin distortion 1	136432
codecheck/outputs/fig5bii.pdf	Fig 5b - PSDM rod and melanopsin distortion 2	134561
codecheck/outputs/fig5ci.pdf	Fig 5c - PSDM rod and melanopsin distortion 1	131587
codecheck/outputs/fig5cii.pdf	Fig 5c - PSDM rod and melanopsin distortion 2	127251
codecheck/outputs/fig6a_box_ CRT.pdf	Fig 6a - PSDM box plot CRT	52595
codecheck/outputs/fig6a_box_	Fig 6a - PSDM box plot DP	53363

CODECHECKER notes

The GitHub repo This check is based on the commit `4e82cf6eebf5a7c79a41fe7706654b6d33cb3418`. Code was written in Matlab. I went through the following steps: 1. Installing Matlab 2. Running `main()` 3. Installing Statistics and Machine Learning Toolbox 4. Running `main()` 5. Installing Image Processing Toolbox 6. Running `main()`

Running `main()` took approx. 2 minutes to complete my laptop.

Recommendations

I suggest to the authors to consider the following suggestions for their next publication or workflow:

- Also document the free toolboxes used

These recommendations were already applied at the time of writing this report.

Manifest files

fig1a.pdf

Comment: Fig 1 - spectral sensitivities 1

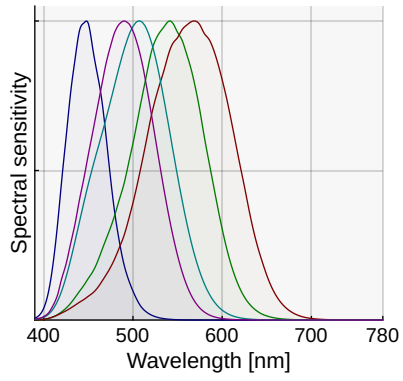


fig1b.pdf

Comment: Fig 1 - example signals 1

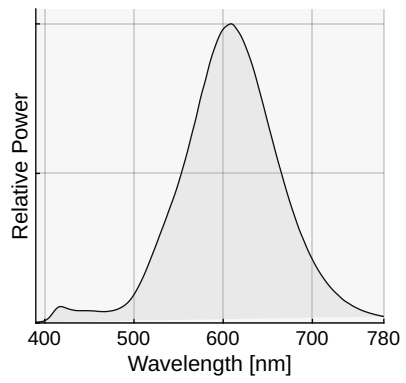


fig1c.pdf

Comment: Fig 1 - spectral sensitivities 2

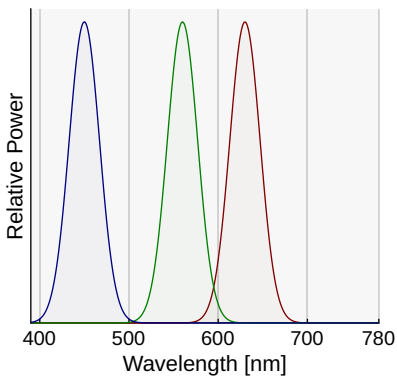


fig1d.pdf

Comment: Fig 1 - example signals 2

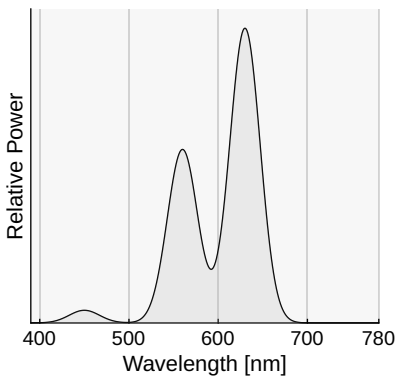


fig1e_inserted.pdf

Comment: Fig 1 - inset of contrast for R and I

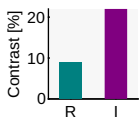


fig1e_inserted.pdf

Comment: Fig 1 - response bar chart 1

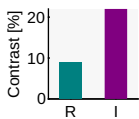


fig1e.pdf

Comment: Fig 1 - spectral sensitivities 3

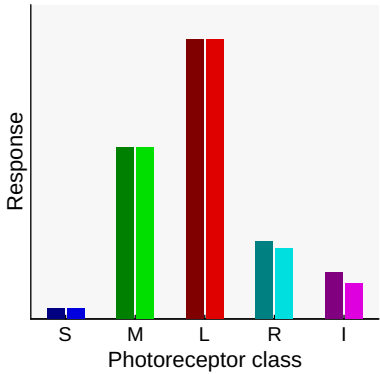


fig1g.pdf

Comment: Fig 1 - example signals 3

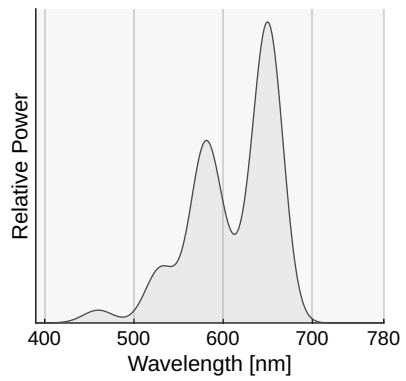


fig1h.pdf

Comment: Fig 1 - response bar chart 2

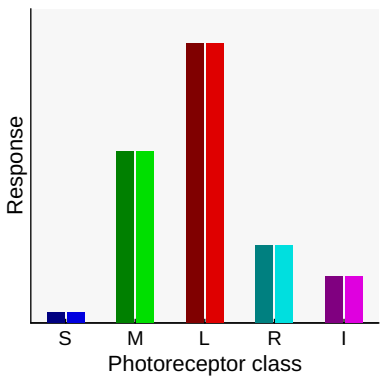


fig2a.pdf

Comment: Fig 2 - equal-luminance MacLeod-Boynton locus 1

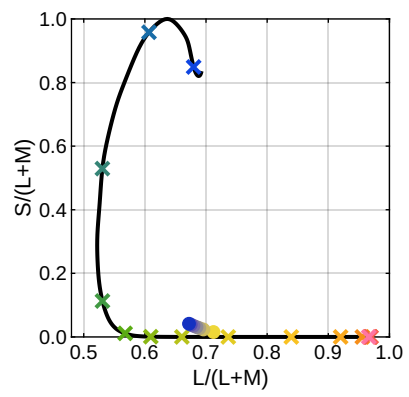


fig2a_insert.pdf

Comment: Fig 2 - equal-luminance MacLeod-Boynton locus inset 1

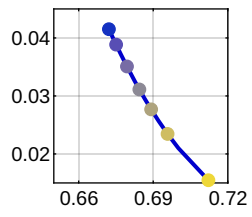


fig2b.pdf

Comment: Fig 2 - equal-luminance MacLeod-Boynton locus 2

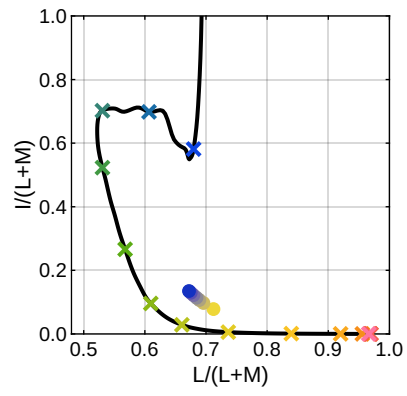


fig2b_insert.pdf

Comment: Fig 2 - equal-luminance MacLeod-Boynton locus inset 2

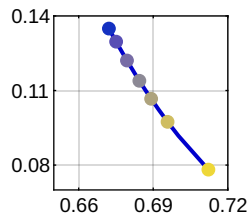


fig2c.pdf

Comment: Fig 2 - equal-luminance MacLeod-Boynton locus 3

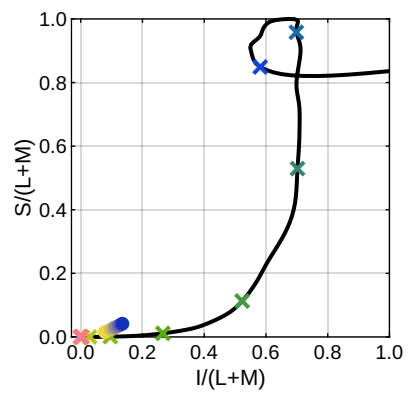


fig2c_insert.pdf

Comment: Fig 2 - equal-luminance MacLeod-Boynton locus inset 3

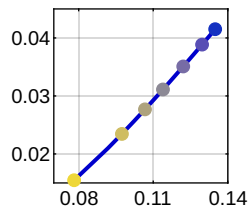


fig3a.pdf

Comment: Fig 3a - display primaries

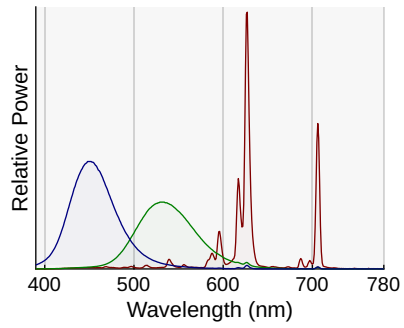


fig3b.pdf

Comment: Fig 3b - display primaries

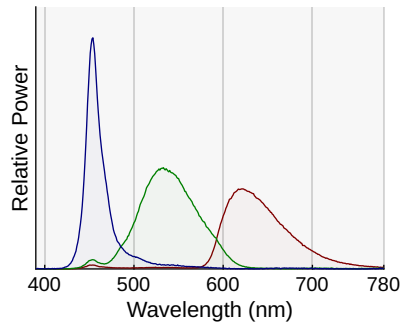


fig3c.pdf

Comment: Fig 3c - display primaries

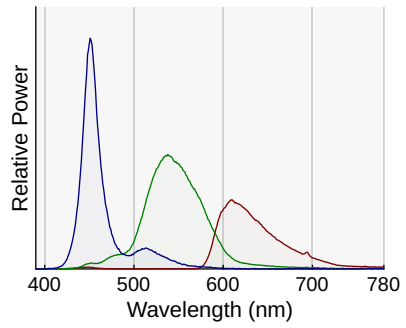


fig3d.pdf

Comment: Fig 3d - display primaries

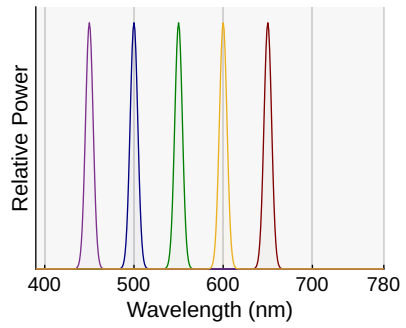


fig3e.pdf

Comment: Fig 3e - display primaries

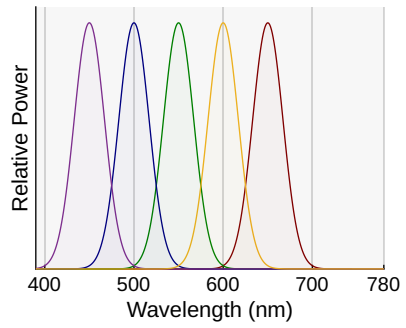


fig3f.pdf

Comment: Fig 3f - display primaries

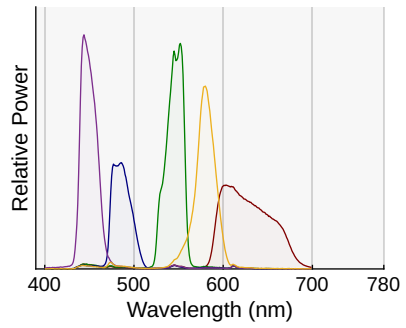


fig3g.pdf

Comment: Fig 3g - display primaries

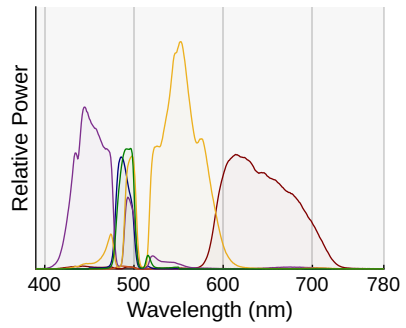


fig3h.pdf

Comment: Fig 3h - display primaries

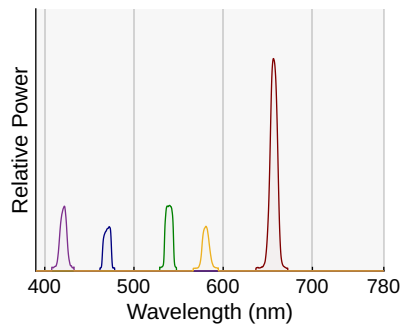


fig4a.png

Comment: Fig 4a - chromaticity reproduction diagram

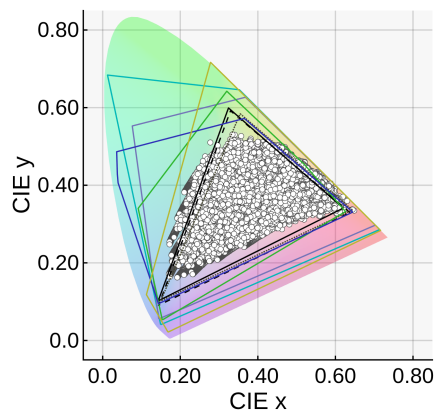


fig4b.pdf

Comment: Fig 4b - chromaticity reproduction bar graph

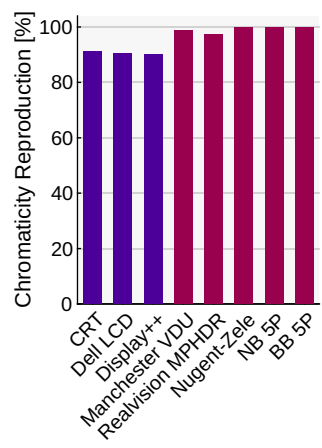


fig4c.pdf

Comment: Fig 4c - PSRM (photoreceptor signal reproduction) bar graph

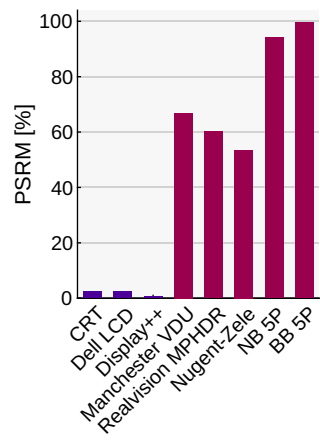


fig5ai.pdf

Comment: Fig 5a - PSDM rod and melanopsin distortion 1

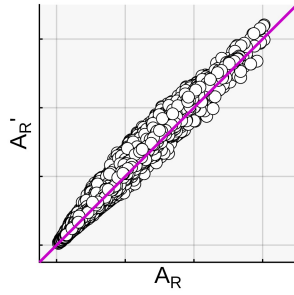


fig5a.ii.pdf

Comment: Fig 5a - PSDM rod and melanopsin distortion 2

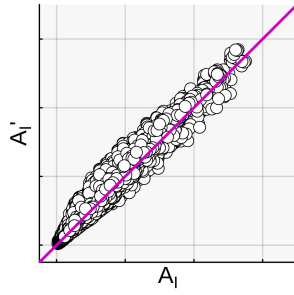


fig5bi.pdf

Comment: Fig 5b - PSDM rod and melanopsin distortion 1

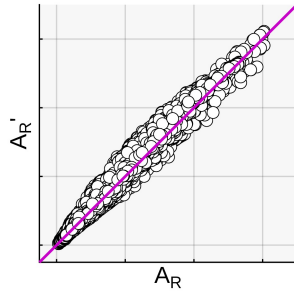


fig5bii.pdf

Comment: Fig 5b - PSDM rod and melanopsin distortion 2

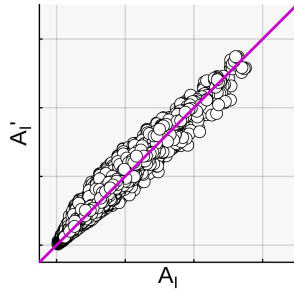


fig5ci.pdf

Comment: Fig 5c - PSDM rod and melanopsin distortion 1

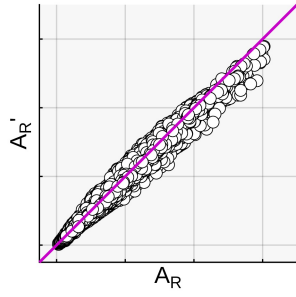


fig5cii.pdf

Comment: Fig 5c - PSDM rod and melanopsin distortion 2

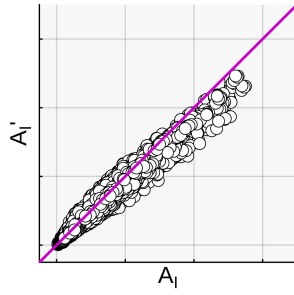


fig6a_box_CRT.pdf

Comment: Fig 6a - PSDM box plot CRT

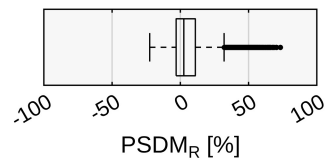


fig6a_box_DP.pdf

Comment: Fig 6a - PSDM box plot DP

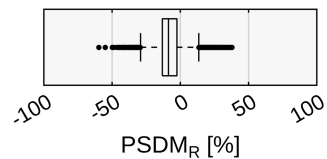


fig6a_box_LCD.pdf

Comment: Fig 6a - PSDM box plot LCD

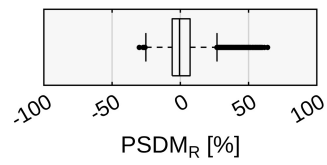


fig6a_xy_CRT.pdf

Comment: Fig 6a - PSDM xy plot CRT

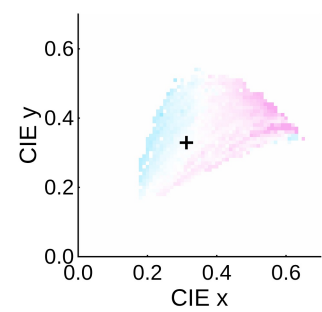


fig6a_xy_DP.pdf

Comment: Fig 6a - PSDM xy plot DP

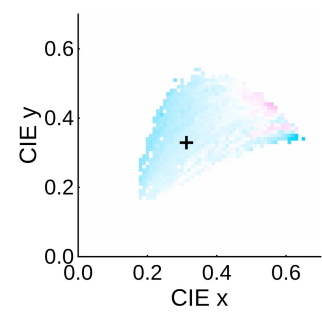


fig6a_xy_LCD.pdf

Comment: Fig 6a - PSDM xy plot LCD

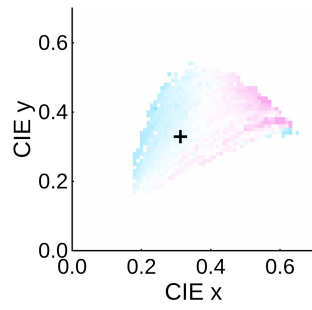


fig6b_box_CRT.pdf

Comment: Fig 6b - PSDM box plot CRT

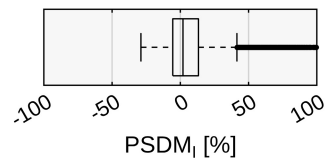


fig6b_box_DP.pdf

Comment: Fig 6b - PSDM box plot DP

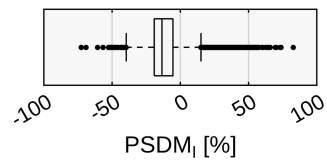


fig6b_box_LCD.pdf

Comment: Fig 6b - PSDM box plot LCD

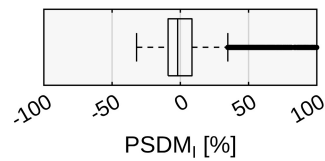


fig6b_xy_CRT.pdf

Comment: Fig 6b - PSDM xy plot CRT

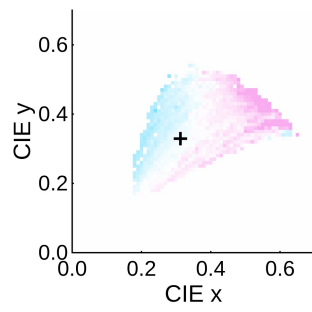


fig6b_xy_DP.pdf

Comment: Fig 6b - PSDM xy plot DP

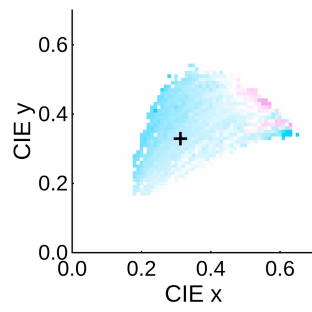


fig6b_xy_LCD.pdf

Comment: Fig 6b - PSDM xy plot LCD

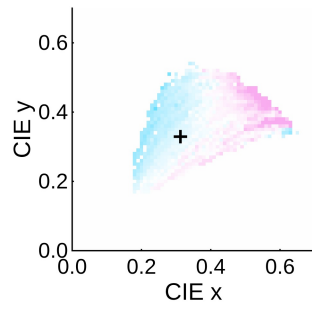


fig7_LvsS_all.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, full database

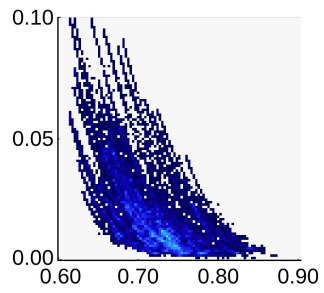


fig7_LvsS_crt.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, CRT

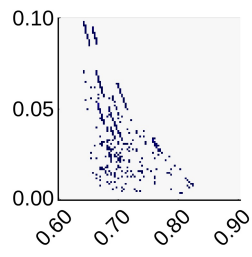


fig7_LvsS_dp.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, DP

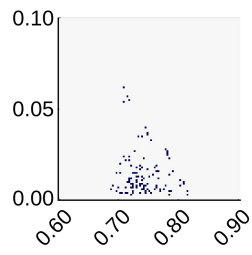


fig7_LvsS_lcd.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, LCD

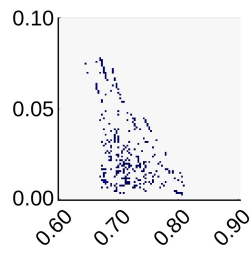


fig7_Lvsl_all.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, full database

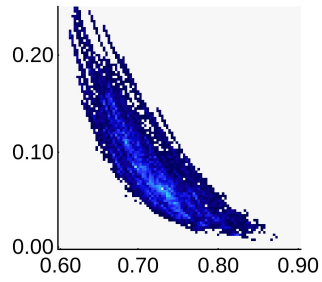


fig7_Lvsl_crt.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, CRT

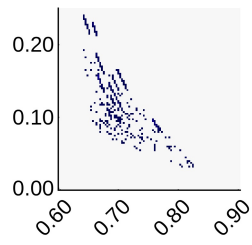


fig7_Lvsl_dp.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, DP

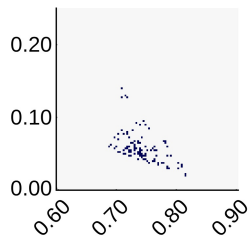


fig7_Lvsl_lcd.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, LCD

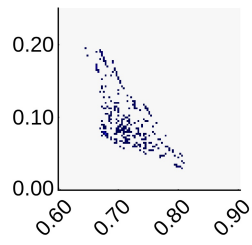


fig7_lvsS_all.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, full database

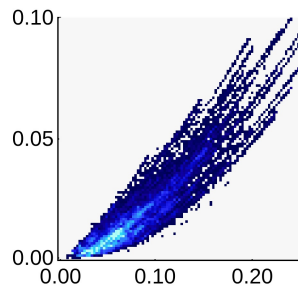


fig7_lvsS_crt.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, CRT

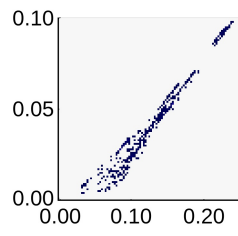


fig7_lvsS_dp.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, DP

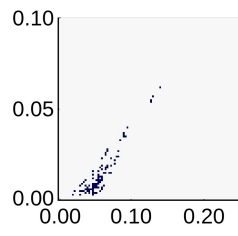


fig7_lvsS_lcd.pdf

Comment: Fig 7 - equal-luminance excitation diagrams, LCD

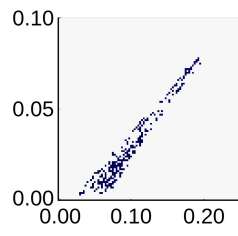


fig8_LvsS_MPHDR.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, MPHDR

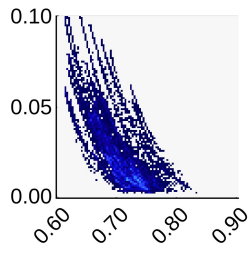


fig8_LvsS_Man.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, Man

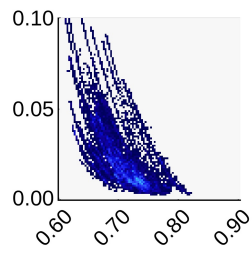


fig8_LvsS_nb5p.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, np5p

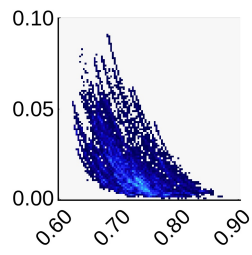


fig8_LvsS_bb5p.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, bb5p

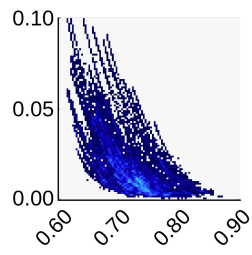


fig8_lvsS_NZ.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, NZ

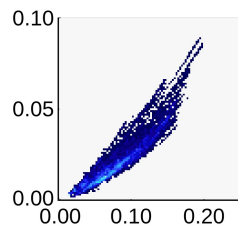


fig8_lvsS_MPHDR.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, MPHDR

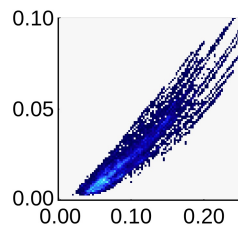


fig8_lvsS_Man.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, Man

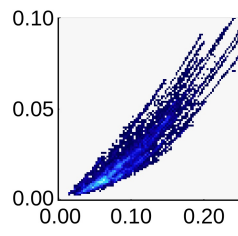


fig8_lvsS_nb5p.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, np5p

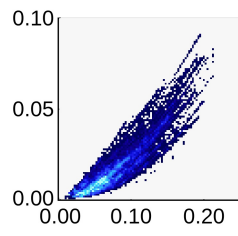


fig8_lvsS_bb5p.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, bb5p

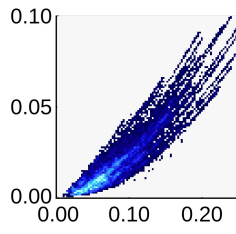


fig8_lvsS_NZ.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, NZ

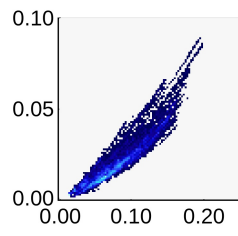


fig8_Lvsl_MPHDR.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, MPHDR

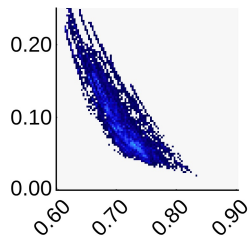


fig8_Lvsl_Man.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, Man

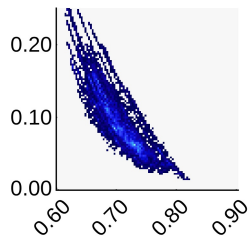


fig8_Lvsl_nb5p.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, np5p

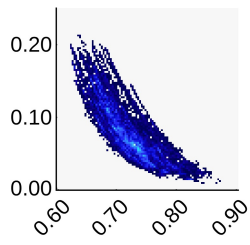


fig8_Lvsl_bb5p.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, bb5p

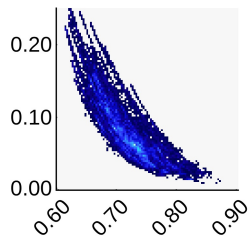


fig8_Lvsl_NZ.pdf

Comment: Fig 8 - equal-luminance excitation diagrams, NZ

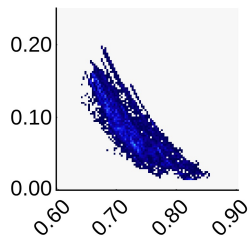


fig9a.pdf

Comment: Fig 9 - bit depth

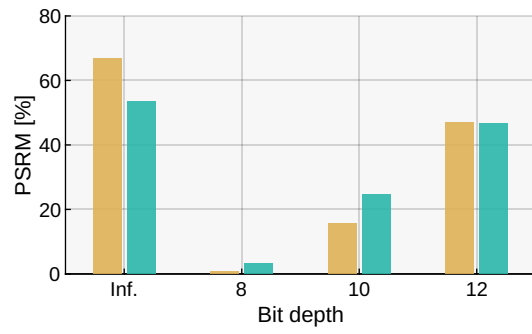
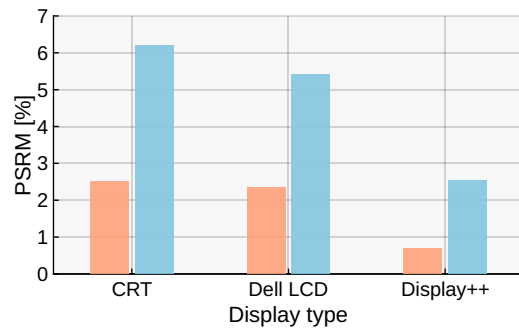


fig9b.pdf

Comment: Fig 9 - tolerance analysis



Acknowledgements

I would like to thank Takuma Morimoto for promptly addressing any upcoming issues.

Citing this document

Simon Christ (29.0). CODECHECK Certificate 2026-024. Zenodo. 10.5281/zenodo.23166593

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```
sessionInfo()
```

```
R version 4.5.2 (2025-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 26.04.1 LTS
```

```
Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.1
LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.1; LAPACK version 3.12.0
```

```
locale:
 [1] LC_CTYPE=de_DE.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=de_DE.UTF-8      LC_COLLATE=de_DE.UTF-8
 [5] LC_MONETARY=de_DE.UTF-8  LC_MESSAGES=de_DE.UTF-8
 [7] LC_PAPER=de_DE.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
```

```
[11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
```

```
time zone: Europe/Berlin
```

```
tzcode source: system (glibc)
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  
[6] methods    base
```

```
other attached packages:
```

```
[1] tibble_3.3.1      xtable_1.8-8  
[3] yaml_2.3.12       rprojroot_2.1.1  
[5] knitr_1.52        codecheck_0.31.0.9000  
[7] parsedate_1.3.2   R.cache_0.17.0  
[9] gh_1.6.1
```

```
loaded via a namespace (and not attached):
```

```
[1] utf8_1.2.6      generics_0.1.4  xml2_1.6.0  
[4] zen4R_0.10.6    stringi_1.8.9   httpcode_0.3.0  
[7] digest_0.6.39   magrittr_2.0.5  evaluate_1.0.5  
[10] fastmap_1.2.0   plyr_1.8.9      cellranger_1.1.0  
[13] R.oo_1.27.1     jsonlite_2.0.0  R.utils_2.13.0  
[16] whisker_0.4.1   crul_1.6.0      httr_1.4.9  
[19] purrr_1.2.2     XML_3.99-0.25   cli_3.6.6  
[22] pdftools_3.9.1  rlang_1.3.0     R.methodsS3_1.8.2  
[25] cachem_1.1.0    otel_0.2.0      tools_4.5.2  
[28] memoise_2.0.1   dplyr_1.2.1     curl_8.0.0  
[31] vctr_0.7.3      R6_2.6.1        lifecycle_1.0.5  
[34] magick_2.9.1    git2r_0.36.2    stringr_1.6.0  
[37] rsvg_2.7.0      fs_2.1.0        pkgconfig_2.0.3  
[40] rorcid_0.7.0    osfr_0.2.9      pillar_1.11.1  
[43] glue_1.8.1      Rcpp_1.1.2      xfun_0.61  
[46] tidyselect_1.2.1 keyring_1.4.1    htmltools_0.5.9  
[49] rmarkdown_2.32  qpdf_1.4.1      compiler_4.5.2  
[52] askpass_1.2.1   fauxpas_0.6.0   readxl_1.5.0.1
```